

Tutorial 01 — From a Table of Counts to a Denoiser

Mathematical Foundations of Diffusion Models
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Practice sheet — worked derivations

The spine of this tutorial

HW 1 ends by building a denoiser. Nobody told you that.

Problem 5 has a hidden cause Z , a noisy observation X , a prior, a likelihood, a posterior $\mathbb{P}(Z \mid X)$, and a conditional mean $\mathbb{E}[Z \mid X]$. That is, structurally and completely, what a diffusion model's denoiser computes. The only difference is that the homework's hidden cause takes two values and its observation takes three, so the whole thing fits in a 2×3 table.

This tutorial makes that sentence literal. Part A walks HW 1 as derivations rather than answers, building toward Problem 5. Part B keeps the structure and replaces the table with a Gaussian — at which point the posterior becomes a logistic function, the conditional mean becomes $\tanh$, and we have derived the denoiser of note 01 with no new ideas at all.

Along the way Part B installs the two other computations that Module 3 runs on: the Gaussian composition rule that produces $q(x_t \mid x_0)$ in closed form, and the KL-between-Gaussians identity that turns the DDPM objective into a squared error.

Part A — HW 1, worked as derivations

1. Problem 1: a random variable is a function

The concept

$X : \Omega \rightarrow \mathbb{R}$ is an ordinary deterministic function. It is not random and it is not a variable. **All the randomness sits in which ω turns up**, and the function merely reports a number about it.

(a) $\Omega = \{1, 2, 3, 4, 5, 6\}$. An event is *any* subset, and a subset is built by making one independent in/out choice per outcome, so there are $2^6 = 64$ of them.

(b) $Y(\omega) = 1$ exactly when ω is prime:

$$\{\omega : Y(\omega) = 1\} = \{2, 3, 5\}, \quad \{\omega : Y(\omega) = 0\} = \{1, 4, 6\}.$$

(c) This is the part worth the time. The claim “ Y maps the event $\{2, 3, 5\}$ to 1” mistakes what Y eats. Its argument is a **single outcome**, never a set:

$$Y(2) = 1, \quad Y(3) = 1, \quad Y(5) = 1$$

— three separate evaluations that happen to return the same number. Nothing is ever evaluated at $\{2, 3, 5\}$. The set arises the other way round, as the **preimage** $Y^{-1}(\{1\})$: it is the *answer* to “where does Y return 1?”, not the *input*. The chain that is actually running:

outcome $\omega \xrightarrow{X}$ number $\longrightarrow$ condition $\longrightarrow$ event (a preimage) $\longrightarrow$ probability

(d) Name the event, then compute.

$$\mathbb{P}(Y = 1) = \mathbb{P}(\{2, 3, 5\}) = \frac{3}{6} = \frac{1}{2} = 0.500$$

$$\{X \leq 3\} \cap \{Y = 1\} = \{1, 2, 3\} \cap \{2, 3, 5\} = \{2, 3\}, \quad \mathbb{P} = \frac{2}{6} = \frac{1}{3} \approx 0.333$$

2. Problem 2: conditioning is division by a margin

The concept

Joint, marginal and conditional are three readings of **one** table. The joint is a cell over the net; a marginal sums a row or column; a conditional is a cell over its **row or column total**. Which total you divide by *is* the choice of what you are conditioning on.

Row totals 70, 70, 60; column totals 65, 70, 65; net 200. Both margins total 200 — your own consistency check.

(a) Divide by the net, because we want each group’s share of everything:

$$\mathbb{P}(X = \cdot) = \frac{1}{200}(70, 70, 60) = (0.350, 0.350, 0.300), \quad \mathbb{P}(Y = \cdot) = \frac{1}{200}(65, 70, 65) = (0.325, 0.350, 0.325).$$

(b) Now divide by the **row** total, because conditioning on $X = A$ restricts attention to that row and asks how it is distributed within itself:

$$\mathbb{P}(Y = \cdot \mid X = A) = \frac{1}{70}(40, 20, 10) = \left(\frac{4}{7}, \frac{2}{7}, \frac{1}{7}\right) \approx (0.571, 0.286, 0.143),$$

$$\mathbb{P}(Y = \cdot \mid X = C) = \frac{1}{60}(10, 15, 35) = \left(\frac{1}{6}, \frac{1}{4}, \frac{7}{12}\right) \approx (0.167, 0.250, 0.583).$$

Both sum to 1 — always check. The two rows are drastically different: grade 1 is the most likely outcome from machine A and the *least* likely from machine C . Knowing the machine changes what you expect of the grade, so X and Y are **dependent**.

(c) Conditioning on $Y = 2$ restricts to that **column**, so divide by its total:

$$\mathbb{P}(X = B \mid Y = 2) = \frac{35}{70} = \frac{1}{2} = 0.500.$$

Dividing by 200 instead gives $35/200 = 0.175$, which is the *joint* $\mathbb{P}(X=B, Y=2)$ — a different question, correctly answered.

(d) Under independence the expected count in a cell is row $\times$ column/net:

$$\frac{70 \times 65}{200} = 22.75 \quad \text{against an observed 40.}$$

Far more grade-1 parts come from A than independence would produce, so independence is untenable. Note carefully: agreement in *one* cell would not have established independence — that requires $\mathbb{P}(X=x, Y=y) = \mathbb{P}(X=x)\mathbb{P}(Y=y)$ in **every** cell.

3. Problem 3: Bayes, and why a good test gives a weak posterior

The concept

The posterior weighs the likelihood against the **base rate**. When the base rate is small enough, a large likelihood ratio is still not enough — because the false positives are drawn from a vastly larger pool.

(a) $\mathbb{P}(D) = 0.01$ (prior, unconditional); $\mathbb{P}(+ | D) = 0.95$ (sensitivity); $\mathbb{P}(+ | D^c) = 0.10$ (false-positive rate). The last two are conditionals *pointing in the opposite direction* to the one we want, which is the whole difficulty.

(b) Law of total probability over the partition $\{D, D^c\}$, disjoint and exhaustive by construction:

$$\mathbb{P}(+) = \mathbb{P}(+ | D)\mathbb{P}(D) + \mathbb{P}(+ | D^c)\mathbb{P}(D^c) = (0.95)(0.01) + (0.10)(0.99) = 0.0095 + 0.0990 = 0.1085.$$

Look at the two terms before moving on: 0.0095 from sick students, 0.0990 from healthy ones. **Ten times as many positives come from the healthy group**, and that ratio is the answer to (d) already sitting on the page.

(c) Bayes:

$$\mathbb{P}(D | +) = \frac{\mathbb{P}(+ | D) \mathbb{P}(D)}{\mathbb{P}(+)} = \frac{0.0095}{0.1085} = \frac{19}{217} \approx 0.0876.$$

(d) The condition is rare. In 10,000 screened students only 100 have it, of whom 95 test positive — but 990 of the 9,900 healthy students also test positive. True positives are outnumbered roughly 10:1.

Which number to change: the **false-positive rate**, downward. It is what generates the 990, and it dominates the denominator.

change	$\mathbb{P}(D +)$
as given	0.0876
sensitivity $0.95 \rightarrow 1.00$ (the best possible)	0.0917
false-positive rate $0.10 \rightarrow 0.05$	$95/(95+495) \approx 0.161$

Perfecting the sensitivity buys almost nothing; halving the false-positive rate nearly doubles the posterior.

The natural-frequency check, which makes the result obvious rather than merely correct:

	tests +	tests −	total
has condition	95	5	100
does not	990	8,910	9,900
total	1,085	8,915	10,000

$\mathbb{P}(D | +) = 95/1085 = 19/217$, identical. And note the table has turned Problem 3 into Problem 2 — same object, different presentation.

4. Problem 4: the continuous analogue

The concept

Everything in Problem 2 survives the move to densities with sums replaced by integrals. Marginalise = integrate the other variable out. Condition = divide by the margin. The normalising constant is the analogue of the net total.

(a) The constant makes the density integrate to 1 over its region — and the region is $[0, 1] \times [0, 2]$, not the unit square:

$$\int_0^2 \int_0^1 (x + y) dx dy = \int_0^2 \left(\frac{1}{2} + y \right) dy = \left[\frac{y}{2} + \frac{y^2}{2} \right]_0^2 = 1 + 2 = 3 \implies c = \frac{1}{3}.$$

(b) Integrate out the *other* variable, and carry the ranges:

$$f_X(x) = \int_0^2 \frac{1}{3}(x + y) dy = \frac{1}{3}(2x + 2) = \frac{2}{3}(x + 1), \quad 0 \leq x \leq 1,$$

$$f_Y(y) = \int_0^1 \frac{1}{3}(x + y) dx = \frac{1}{3} \left(\frac{1}{2} + y \right), \quad 0 \leq y \leq 2.$$

Checks: $\int_0^1 \frac{2}{3}(x + 1) dx = \frac{2}{3} \cdot \frac{3}{2} = 1$ and $\int_0^2 \frac{1}{3} \left(\frac{1}{2} + y \right) dy = \frac{1}{3}(1 + 2) = 1$.

(c) Divide the joint by the margin:

$$f(y | x) = \frac{f(x, y)}{f_X(x)} = \frac{\frac{1}{3}(x + y)}{\frac{2}{3}(x + 1)} = \frac{x + y}{2(x + 1)}, \quad 0 \leq y \leq 2.$$

Check: $\int_0^2 \frac{x + y}{2(x + 1)} dy = \frac{2x + 2}{2(x + 1)} = 1$ for **every** x — as it must be, since a conditional density is a genuine density in y for each fixed x .

What was divided by: $f_X(x)$ is the total mass along the vertical slice at that x . This is exactly dividing a row of a count table by its row total, and the whole of Problem 2(b) is the discrete shadow of this line.

(d) Not independent. Compare explicitly:

$$f_X(x) f_Y(y) - f(x, y) = \frac{2}{9}(x + 1) \left(\frac{1}{2} + y \right) - \frac{1}{3}(x + y) = \frac{1}{9}(y - 1)(2x - 1),$$

non-zero except on the lines $x = \frac{1}{2}$ and $y = 1$. Equivalently and faster: $x + y$ does not factor as (a function of x) $\times$ (a function of y), and a joint density factorises **if and only if** the variables are independent.

5. Problem 5: a hidden cause — the discrete denoiser

The concept

The whole tutorial in miniature. A cause Z you cannot see, an effect X you can, and the machinery that recovers a belief about the first from the second:

$$\text{prior} \times \text{likelihood} \rightarrow \text{joint} \rightarrow \text{marginal} \rightarrow \text{posterior} \rightarrow \text{conditional mean}$$

Problem 2 gave you the joint table for free because both variables were recorded. Here Z is **never observed**, so there is no table to build — only a model, a prior, and Bayes.

(a) Build the joint from $p(z, x) = p(z) p(x | z)$: multiply each row of the likelihood table by its prior.

	$X = 0$	$X = 1$	$X = 2$	row
$Z = 1$ (prior 0.6)	0.42	0.12	0.06	0.60
$Z = 2$ (prior 0.4)	0.08	0.12	0.20	0.40
column	0.50	0.24	0.26	1.00

(b) Sum out Z — the variable you do not observe:

$$\mathbb{P}(X = \cdot) = (0.50, 0.24, 0.26).$$

This is what the inspector actually sees, with the machine identity averaged away. Notice what has been destroyed: many different machine-mixtures produce this same marginal, so from $p(x)$ alone the split cannot be recovered. **Marginalising is lossy, and the lost thing is the decomposition.**

(c) Bayes, with the column of the joint table already serving as the numerator:

$$\mathbb{P}(Z = 1 \mid X = x) = \frac{\overbrace{\mathbb{P}(X = x \mid Z = 1)}^{\text{likelihood}} \overbrace{\mathbb{P}(Z = 1)}^{\text{prior}}}{\underbrace{\mathbb{P}(X = x)}_{\text{evidence}}}$$

x	computation	$\mathbb{P}(Z = 1 \mid X = x)$
0	0.42/0.50	21/25 = 0.840
1	0.12/0.24	1/2 = 0.500
2	0.06/0.26	3/13 $\approx$ 0.231

Read the trend: zero defects is evidence for the older machine, two defects is evidence against it, and one defect is exactly uninformative — it moves the belief from the prior 0.6 to 0.5, which is a *change*, so even the “uninformative” reading carries information.

(d) Z takes values in $\{1, 2\}$, so with $p = \mathbb{P}(Z = 1 \mid X = x)$ the average is taken *over the posterior of Z at that fixed x* :

$$\mathbb{E}_{Z \mid X=x}[Z] = 1 \cdot p + 2 \cdot (1 - p) = 2 - p.$$

x	$\mathbb{E}_{Z \mid X=x}[Z]$
0	29/25 = 1.160
1	3/2 = 1.500
2	23/13 $\approx$ 1.769

These are not in $\{1, 2\}$ and need not be. **A conditional expectation is a weighted average of the possible values, not a choice among them.** 1.16 says “probably machine 1, with some chance of machine 2” — it summarises a distribution rather than naming a label. If forced to name one, report the posterior *mode*, not the mean.

(e) The tower property, verified. Two *different* averages are involved, and the subscripts say which is which: the inner one runs over Z with x fixed (that is column (d)), and the outer one runs over X , weighting by how often each x occurs:

$$\mathbb{E}_X \left[\mathbb{E}_{Z \mid X}[Z \mid X] \right] = \underbrace{(0.50)}_{\mathbb{P}(X=0)}(1.160) + \underbrace{(0.24)}_{\mathbb{P}(X=1)}(1.500) + \underbrace{(0.26)}_{\mathbb{P}(X=2)}(1.769) = 0.58 + 0.36 + 0.46 = 1.40,$$

against $\mathbb{E}_Z[Z] = 1(0.6) + 2(0.4) = 1.40$. Equal exactly, not approximately.

Note what the inner object is: $\mathbb{E}_{Z|X}[Z | X]$ is not one number but a **random variable**, taking the values 1.160, 1.500, 1.769 with probabilities 0.50, 0.24, 0.26. The outer $\mathbb{E}_X$ is what averages it. §7.0 returns to this distinction, which is the one that makes the tower property look circular when the subscripts are dropped.

In words: averaging the conditional averages, **weighted by how often each condition occurs**, returns the overall average. Observing X redistributes your estimate of Z but cannot move it on average — the precise sense in which an observation is informative about *cases* without being biased overall.

Part B — the three computations Module 3 runs on

The concept

Part A’s last problem had a discrete hidden cause and a discrete observation, recovered by Bayes. Real data is continuous and the noise is Gaussian, so three things must be installed before the same pipeline can run:

- §6 — what a Gaussian does under scaling and addition, so that “add noise t times” collapses to a single formula;
- §7 — that a conditional mean is what squared-error training produces, and what it looks like when the table becomes a Gaussian;
- §8 — how to measure the gap between two Gaussians, which turns out to be a squared error.

Together they are essentially all the mathematics in a DDPM.

6. Gaussian algebra, ending in the composition rule

6.1 Affine maps

Claim. If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $a \neq 0$, then $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.

Derivation by change of variables (take $a > 0$; $a < 0$ differs only by $|a|$). Start from the CDF, because that is where a monotone map is easiest:

$$F_Y(y) = \mathbb{P}(aX + b \leq y) = \mathbb{P}\left(X \leq \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right).$$

Differentiate, picking up $1/a$ from the chain rule:

$$f_Y(y) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right) = \frac{1}{a\sqrt{2\pi}\sigma} \exp\left(-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right).$$

Now clear the inner fraction — the only step with any content:

$$\frac{\left(\frac{y-b}{a} - \mu\right)^2}{\sigma^2} = \frac{(y-b-a\mu)^2}{a^2\sigma^2},$$

so f_Y is a Gaussian density with mean $a\mu + b$ and variance $a^2\sigma^2$. ■

The consequence used constantly: $X = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0, 1)$. Every Gaussian is standard noise, scaled and shifted — the *reparametrisation trick*, and the reason a sampler only ever needs $\mathcal{N}(0, 1)$.

6.2 Sums of independent Gaussians

Claim. $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ independent $\Rightarrow X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Derivation by MGF, the short route, set up in the W03 deck. For independent variables the MGF of a sum is the product of the MGFs:

$$M_{X+Y}(t) = M_X(t) M_Y(t) = e^{\mu_1 t + \frac{1}{2}\sigma_1^2 t^2} e^{\mu_2 t + \frac{1}{2}\sigma_2^2 t^2} = e^{(\mu_1 + \mu_2)t + \frac{1}{2}(\sigma_1^2 + \sigma_2^2)t^2}.$$

That is the Gaussian MGF with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$; by the uniqueness theorem the distribution is that Gaussian. ■

6.3 The composition — the payoff

The single most load-bearing computation in the course, and it is two lines.

Setup. One noising step, with x_0 held fixed and $\varepsilon \sim \mathcal{N}(0, 1)$:

$$x_1 = \sqrt{\alpha_1} x_0 + \sqrt{1 - \alpha_1} \varepsilon_1, \quad x_2 = \sqrt{\alpha_2} x_1 + \sqrt{1 - \alpha_2} \varepsilon_2,$$

with $\varepsilon_1, \varepsilon_2$ independent. **What is x_2 in terms of x_0 ?**

Substitute the first into the second:

$$\begin{aligned} x_2 &= \sqrt{\alpha_2} \left(\sqrt{\alpha_1} x_0 + \sqrt{1 - \alpha_1} \varepsilon_1 \right) + \sqrt{1 - \alpha_2} \varepsilon_2 \\ x_2 &= \sqrt{\alpha_1 \alpha_2} x_0 + \underbrace{\sqrt{\alpha_2(1 - \alpha_1)} \varepsilon_1 + \sqrt{1 - \alpha_2} \varepsilon_2}_{\text{two independent Gaussians, mean 0}} \end{aligned}$$

Add the variances, by §6.1 for the scalings and §6.2 for the sum:

$$\alpha_2(1 - \alpha_1) + (1 - \alpha_2) = \alpha_2 - \alpha_1 \alpha_2 + 1 - \alpha_2 = 1 - \alpha_1 \alpha_2.$$

The α_2 terms cancel exactly, and therefore

$$\boxed{x_2 = \sqrt{\alpha_1 \alpha_2} x_0 + \sqrt{1 - \alpha_1 \alpha_2} \bar{\varepsilon}, \quad \bar{\varepsilon} \sim \mathcal{N}(0, 1).}$$

Two noising steps have the form of one noising step, with $\alpha_1 \alpha_2$ in place of α . By induction, writing $\bar{\alpha}_t = \prod_{s \leq t} \alpha_s$:

$$\boxed{x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \text{i.e.} \quad q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I).$$

The concept

Three things to draw out of that formula.

The parametrisation was designed for this. The pair $(\sqrt{\alpha}, \sqrt{1 - \alpha})$ is chosen precisely so the noise variances telescope to $1 - \bar{\alpha}_t$. Any other choice still gives a Gaussian; this one gives a *tidy* Gaussian.

Total variance is preserved. If x_0 has unit variance then x_t has variance $\bar{\alpha}_t + (1 - \bar{\alpha}_t) = 1$ at every t . The process trades signal for noise at a fixed total — hence “variance preserving”.

The destination is free. $\alpha_s < 1$ for each s , so $\bar{\alpha}_t \rightarrow 0$ and $x_t \rightarrow \mathcal{N}(0, I)$. The forward process needs no training and its endpoint is a distribution we can sample perfectly. *That* is why diffusion models start by destroying the data.

7. Conditional expectation is what squared error computes

7.0 Notation: which distribution is each $\mathbb{E}$ taken over?

Every expectation below carries a subscript naming the distribution it integrates against. Without them the tower property looks circular — it appears to say $\mathbb{E}[\mathbb{E}[\cdot]] = \mathbb{E}[\cdot]$, which is either trivial or meaningless. It is neither, because *the three $\mathbb{E}$ ’s are three different operators*.

symbol	integrates over	the result is
$\mathbb{E}_{Y X=x}[\cdot]$	Y , with x held <i>fixed</i>	a number , depending on x
$\mathbb{E}_{Y X}[\cdot X]$	Y , with X left <i>symbolic</i>	a random variable : a function of X
$\mathbb{E}_X[\cdot]$	X	a number
$\mathbb{E}_{(X,Y)}[\cdot]$	both, jointly	a number

The concept

The one distinction that clears up the confusion.

$$\underbrace{\mathbb{E}_{Y|X=x}[Y]}_{\text{a number}} \quad \text{versus} \quad \underbrace{\mathbb{E}_{Y|X}[Y | X]}_{\text{a random variable}}$$

The second is what you get by *not* plugging in a value: it is the function $x \mapsto \mathbb{E}_{Y|X=x}[Y]$, regarded as a random quantity because its input X is random. In HW 1 Problem 5 the three numbers 1.160, 1.500, 1.769 are the first object evaluated at $x = 0, 1, 2$; the second object is the whole table, which takes those three values with probabilities 0.50, 0.24, 0.26.

The tower property then says something with content: averaging that random variable over X returns the plain average of Y ,

$$\mathbb{E}_X \left[\mathbb{E}_{Y|X}[Y | X] \right] = \mathbb{E}_Y[Y],$$

and more generally, for any h ,

$$\mathbb{E}_{(X,Y)}[h(X,Y)] = \mathbb{E}_X \left[\mathbb{E}_{Y|X}[h(X,Y) | X] \right].$$

Outer operator over X , inner operator over Y given X . They are not the same $\mathbb{E}$, which is why the statement is not a tautology.

Two facts used repeatedly below, both immediate once the subscripts are visible:

- **Pull-out.** If $a(X)$ depends on X alone, it is a *constant* for the inner operator: $\mathbb{E}_{Y|X}[a(X)W | X] = a(X)\mathbb{E}_{Y|X}[W | X]$.
- **Collapse.** If the integrand depends on X alone, the joint expectation is just the X -expectation: $\mathbb{E}_{(X,Y)}[a(X)] = \mathbb{E}_X[a(X)]$.

7.1 The best-predictor theorem

Claim. Among *all* functions g , the one minimising $\mathbb{E}_{(X,Y)}[(Y - g(X))^2]$ is $g(X) = \mathbb{E}_{Y|X}[Y | X]$.

Derivation. Define the function

$$m(x) = \mathbb{E}_{Y|X=x}[Y], \quad \text{so that} \quad m(X) = \mathbb{E}_{Y|X}[Y | X]$$

is a random variable depending on X only. Split the error around it:

$$Y - g(X) = (Y - m(X)) + (m(X) - g(X)).$$

Square and take the *joint* expectation — both terms involve X and Y , so this is $\mathbb{E}_{(X,Y)}$ throughout:

$$\mathbb{E}_{(X,Y)}[(Y - g)^2] = \mathbb{E}_{(X,Y)}[(Y - m)^2] + 2\mathbb{E}_{(X,Y)}[(Y - m)(m - g)] + \mathbb{E}_{(X,Y)}[(m - g)^2].$$

The cross term vanishes, and this is the only step that does any work. Apply the tower property to *split the joint expectation into two*:

$$\mathbb{E}_{(X,Y)}[(Y - m)(m - g)] = \mathbb{E}_X \left[\mathbb{E}_{Y|X}[(Y - m(X))(m(X) - g(X)) | X] \right].$$

Work on the *inner* operator only. It integrates over Y with X held fixed, so $m(X) - g(X)$ is a constant for it and pulls out:

$$\mathbb{E}_{Y|X}[(Y - m(X))(m(X) - g(X)) | X] = \underbrace{(m(X) - g(X))}_{\text{constant for } \mathbb{E}_{Y|X}} \underbrace{\mathbb{E}_{Y|X}[Y - m(X) | X]}_{= m(X) - m(X) = 0} = 0.$$

The inner expectation is zero *for every value of* X , so the outer $\mathbb{E}_X$ of it is zero as well. The third term involves X alone, so it collapses to $\mathbb{E}_X$, and the decomposition reads

$$\mathbb{E}_{(X,Y)}[(Y - g)^2] = \underbrace{\mathbb{E}_{(X,Y)}[(Y - m)^2]}_{\text{does not involve } g} + \underbrace{\mathbb{E}_X[(m(X) - g(X))^2]}_{\geq 0, \text{ zero only at } g=m}.$$

The first term is a floor you cannot get below — the noise in Y that X simply does not determine. The second is entirely under your control and is minimised, uniquely, at $g = m$. ■

The concept

Least squares does not fit a curve. It computes a conditional expectation.

Anything trained with a squared-error loss is estimating $\mathbb{E}_{\text{target}|\text{input}}[\text{target}]$ — whether or not the person who wrote the loss intended it. This is why a diffusion model trained on $\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2$ ends up computing a posterior mean without any Bayesian machinery appearing in the code.

Free corollary — the law of total variance. Put $g \equiv \mathbb{E}_Y[Y]$, a constant function, into the same identity:

$$\underbrace{\mathbb{E}_Y[(Y - \mathbb{E}_Y Y)^2]}_{\text{Var}_Y(Y)} = \underbrace{\mathbb{E}_{(X,Y)}[(Y - m(X))^2]}_{\mathbb{E}_X[\text{Var}_{Y|X}(Y|X)]} + \underbrace{\mathbb{E}_X[(m(X) - \mathbb{E}_Y Y)^2]}_{\text{Var}_X(\mathbb{E}_{Y|X}[Y|X])}.$$

Read the two right-hand terms with their subscripts and the statement becomes concrete: the first averages, *over* X , the spread of Y that remains once X is known; the second measures how much the conditional mean itself moves *as* X *varies*. Variance splits into what observing X cannot explain plus what it can. One identity, two famous theorems.

7.2 The bridge back to HW 1

In Problem 5 you computed $\mathbb{E}_{Z|X=x}[Z] = 1.160, 1.500, 1.769$.

§7.1 says something stronger than it looked: **a network trained to predict Z from X under squared error would converge to exactly those three numbers**. Not approximately, and not as an artefact of the architecture — they are the minimiser. The homework computed by hand what training computes by descent.

7.3 The same pipeline, with a Gaussian: the tanh denoiser

Now replace the table with a Gaussian and run Problem 5 again.

Setup. The clean signal is $x_0 \in \{-1, +1\}$, each with probability $\frac{1}{2}$ — the two-point dataset of note 01. We observe

$$x_\sigma = x_0 + \sigma \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1),$$

so the likelihood is $p(x_\sigma | x_0) = \mathcal{N}(x_\sigma; x_0, \sigma^2)$. Same pipeline: prior, likelihood, posterior, conditional mean. As in §7.0, the averaging below is over the *posterior* of x_0 at a fixed observed x_σ , which is what the subscript on $\mathbb{E}$ records.

Step 1 — the posterior. Bayes, with the prior $\frac{1}{2}$ cancelling top and bottom:

$$\mathbb{P}(x_0 = +1 | x_\sigma) = \frac{\exp\left(-\frac{(x_\sigma-1)^2}{2\sigma^2}\right)}{\exp\left(-\frac{(x_\sigma-1)^2}{2\sigma^2}\right) + \exp\left(-\frac{(x_\sigma+1)^2}{2\sigma^2}\right)}.$$

Expand the squares, $(x_\sigma \mp 1)^2 = x_\sigma^2 \mp 2x_\sigma + 1$, and notice that $x_\sigma^2 + 1$ appears in **both** exponents. Divide top and bottom by $\exp\left(-\frac{x_\sigma^2+1}{2\sigma^2}\right)$ and it disappears, leaving only the cross terms:

$$\mathbb{P}(x_0 = +1 | x_\sigma) = \frac{e^{x_\sigma/\sigma^2}}{e^{x_\sigma/\sigma^2} + e^{-x_\sigma/\sigma^2}} = \text{logistic}\left(\frac{2x_\sigma}{\sigma^2}\right).$$

Step 2 — the conditional mean. With $p = \mathbb{P}(x_0 = +1 | x_\sigma)$ and values ± 1 , average over the posterior:

$$\mathbb{E}_{x_0|x_\sigma}[x_0] = (+1)p + (-1)(1-p) = 2p - 1 = \frac{e^u - e^{-u}}{e^u + e^{-u}}, \quad u = \frac{x_\sigma}{\sigma^2},$$

which is exactly the definition of the hyperbolic tangent:

$$\boxed{\mathbb{E}_{x_0|x_\sigma}[x_0] = \tanh\left(\frac{x_\sigma}{\sigma^2}\right).}$$

(The literature writes this $\mathbb{E}[x_0 | x_\sigma]$ and leaves the distribution implicit. That is safe *here*, because there is only one thing to average over; it is not safe in §7.1, where two nested operators are in play.)

That is the optimal denoiser for this dataset — the one from note 01 and from the W01 demo, now derived rather than asserted.

Read it against HW 1 Problem 5, line for line:

HW 1, Problem 5	Two-point diffusion
latent $Z \in \{1, 2\}$	latent $x_0 \in \{-1, +1\}$
observed $X \in \{0, 1, 2\}$	observed $x_\sigma \in \mathbb{R}$
prior (0.6, 0.4)	prior $(\frac{1}{2}, \frac{1}{2})$
likelihood: a 2×3 table	likelihood: $\mathcal{N}(x_\sigma; x_0, \sigma^2)$
posterior: three numbers	posterior: $\text{logistic}(2x_\sigma/\sigma^2)$
$\mathbb{E}_{Z X=x}[Z]$: three numbers	$\mathbb{E}_{x_0 x_\sigma}[x_0] = \tanh(x_\sigma/\sigma^2)$

Nothing changed except that the table became a formula.

Step 3 — the two limits, worth a minute each because they say what denoising *is*:

- $\sigma \rightarrow 0$: the argument x_σ/σ^2 blows up, $\tanh$ becomes a step function, and the denoiser confidently returns the nearer of ± 1 . Little noise, so trust the observation.
- $\sigma \rightarrow \infty$: the argument goes to 0 and so does $\tanh$. The denoiser returns 0 — the **prior mean** — because the observation carries nothing. It does not guess; it reports the average.

A denoiser interpolates between trusting the data and trusting the prior, and σ is the dial. Every diffusion model is doing this at every noise level simultaneously.

The concept

Optional — Tweedie’s formula. Stated, not proved:

$$\mathbb{E}_{x_0|x_\sigma}[x_0] = x_\sigma + \sigma^2 \nabla_x \log p_\sigma(x_\sigma).$$

The denoiser and the score are the same object in different clothes, which is the whole content of Module 4. You can check it here directly: differentiate $\log p_\sigma$ for the two-point mixture and confirm the right-hand side collapses to $\tanh(x_\sigma/\sigma^2)$.

8. KL divergence, and why it becomes a squared error

8.1 Definition, and non-negativity

$$\text{KL}(p \| q) = \mathbb{E}_p \left[\log \frac{p(X)}{q(X)} \right] = \int p(x) \log \frac{p(x)}{q(x)} dx.$$

Claim. $\text{KL}(p \| q) \geq 0$, with equality only when $p = q$.

Derivation by Jensen, using that $\log$ is concave so $\mathbb{E}[\log W] \leq \log \mathbb{E}[W]$:

$$-\text{KL}(p \| q) = \mathbb{E}_p \left[\log \frac{q(X)}{p(X)} \right] \leq \log \mathbb{E}_p \left[\frac{q(X)}{p(X)} \right] = \log \int p(x) \frac{q(x)}{p(x)} dx = \log 1 = 0.$$

Equality in Jensen for a strictly concave function requires the random variable to be constant, i.e. q/p constant almost surely — and since both integrate to 1, that constant is 1. ■

8.2 Between two Gaussians, derived

Let $p = \mathcal{N}(\mu_1, \sigma_1^2)$ and $q = \mathcal{N}(\mu_2, \sigma_2^2)$. Take the log-ratio first, since the exponentials make it easy:

$$\log \frac{p(x)}{q(x)} = \log \frac{\sigma_2}{\sigma_1} - \frac{(x - \mu_1)^2}{2\sigma_1^2} + \frac{(x - \mu_2)^2}{2\sigma_2^2}.$$

Now take $\mathbb{E}_p$ term by term. Two expectations are needed, **both under p** , which is where it goes wrong:

$$\mathbb{E}_p[(X - \mu_1)^2] = \sigma_1^2 \quad (\text{the variance of } p, \text{ by definition}),$$

$$\mathbb{E}_p[(X - \mu_2)^2] = \mathbb{E}_p[((X - \mu_1) + (\mu_1 - \mu_2))^2] = \sigma_1^2 + (\mu_1 - \mu_2)^2,$$

the cross term vanishing because $\mathbb{E}_p[X - \mu_1] = 0$. Substituting:

$$\text{KL}(\mathcal{N}(\mu_1, \sigma_1^2) \| \mathcal{N}(\mu_2, \sigma_2^2)) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}.$$

Sanity check: set $\mu_1 = \mu_2$ and $\sigma_1 = \sigma_2$ and every term cancels to 0, as it must.

8.3 The collapse — the line Module 3 depends on

Set the two variances equal, $\sigma_1 = \sigma_2 = \sigma$. The log term is $\log 1 = 0$, and the middle term splits:

$$\frac{\sigma^2 + (\mu_1 - \mu_2)^2}{2\sigma^2} - \frac{1}{2} = \frac{1}{2} + \frac{(\mu_1 - \mu_2)^2}{2\sigma^2} - \frac{1}{2},$$

leaving

$$\boxed{\text{KL}(\mathcal{N}(\mu_1, \sigma^2) \parallel \mathcal{N}(\mu_2, \sigma^2)) = \frac{(\mu_1 - \mu_2)^2}{2\sigma^2}}$$

and in d dimensions with $\Sigma = \sigma^2 I$, the same computation gives $\|\mu_1 - \mu_2\|^2 / 2\sigma^2$.

The concept

A KL divergence between two Gaussians of equal width is a squared error, divided by a constant.

The DDPM training objective is a variational bound that decomposes into a sum of KL terms, one per noise level, each between two Gaussians that share a variance. By the boxed line above, every one of them is a squared error — which is why a derivation full of KL divergences ends at

$$\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2.$$

There is no additional trick in Module 3. There is this identity, applied once per term.

8.4 Maximum likelihood is minimum KL

This closes the loop with HW 2, where you did MLE by hand without being told what it was minimising. Take the average log-likelihood and let n grow; by the law of large numbers,

$$\frac{1}{n} \sum_{i=1}^n \log p_\theta(x_i) \longrightarrow \mathbb{E}_{p_{\text{data}}}[\log p_\theta(X)].$$

Add and subtract $\mathbb{E}_{p_{\text{data}}}[\log p_{\text{data}}]$:

$$\mathbb{E}_{p_{\text{data}}}[\log p_\theta] = \underbrace{-H(p_{\text{data}})}_{\text{no } \theta} - \text{KL}(p_{\text{data}} \parallel p_\theta).$$

The entropy does not involve θ , so

$$\boxed{\arg \max_{\theta} \text{likelihood} = \arg \min_{\theta} \text{KL}(p_{\text{data}} \parallel p_\theta)}.$$

Fitting was never “describe the data”. It was “get close to the data distribution” — we simply never had to name the notion of closeness, because likelihood chose one for us. Naming it is what exposes the choice, and the choice is what the rest of the course is about.

8.5 It is not symmetric, and here are the numbers

“KL is not a distance” is easy to accept verbally and then forget when reasoning. Kill that with arithmetic. Take $p = \mathcal{N}(0, 1)$ and $q = \mathcal{N}(0, \sigma^2)$ in §8.2:

$$\text{KL}(p \parallel q) = \log \sigma + \frac{1}{2\sigma^2} - \frac{1}{2}, \quad \text{KL}(q \parallel p) = -\log \sigma + \frac{\sigma^2}{2} - \frac{1}{2}.$$

At $\sigma = 2$:

direction	value
$\text{KL}(\mathcal{N}(0, 1) \parallel \mathcal{N}(0, 4))$	$\log 2 + \frac{1}{8} - \frac{1}{2} \approx 0.318$
$\text{KL}(\mathcal{N}(0, 4) \parallel \mathcal{N}(0, 1))$	$-\log 2 + 2 - \frac{1}{2} \approx 0.807$

Two and a half times apart, from the same pair of distributions. The asymmetry is not a technicality: $\text{KL}(p \parallel q)$ punishes q for being *narrow* where p has mass (mode-covering), and $\text{KL}(q \parallel p)$ punishes q for being *wide* where p has none (mode-seeking). Which order you write is a modelling decision, and note 09 §6 showed a GAN's practical loss quietly picking the second one.

Closing: what the session installed

Computation	Where it came from	Where it returns
Prior $\times$ likelihood $\rightarrow$ posterior	HW 1 P3, P5	the reverse process; $q(x_{t-1} \mid x_t, x_0)$
Marginalising a latent	HW 1 P5(b)	the intractable $p_\theta(x_0)$; why the ELBO exists
Tower property	HW 1 P5(e)	every expectation argument in Module 3
$\mathbb{E}_{Z \mid X}[Z \mid X]$ as a weighted average	HW 1 P5(d)	the denoiser output
Variances add	§6.2	the $\sqrt{t}$ envelope of the forward process
$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon$	§6.3	training at a random t in one step
Squared error $\Rightarrow$ conditional expectation	§7.1	why the ε -loss is Bayesian
$\mathbb{E}_{x_0 \mid x_\sigma}[x_0] = \tanh(x_\sigma/\sigma^2)$	§7.3	note 01, now derived
KL between equal-variance Gaussians	§8.3	the DDPM objective's final form
MLE = min KL	§8.4	why "which divergence?" organises the field

The concept

The one sentence. A diffusion model is HW 1 Problem 5, run at every noise level at once, with the table replaced by a Gaussian and the posterior mean computed by a network trained under squared error.

Everything in Module 3 is bookkeeping on top of that sentence.

Follow-on exercises

- Redo HW 1 Problem 5 with prior $\mathbb{P}(Z=1) = 0.9$. Which posterior moves most, and why is it the one for $x = 2$?
- Show $\mathbb{E}_X[\text{Var}_{Y \mid X}(Y \mid X)] \leq \text{Var}_Y(Y)$, and say when equality holds. (*What does equality mean about X ?*)
- Three noising steps: show directly that $x_3 = \sqrt{\alpha_1\alpha_2\alpha_3}x_0 + \sqrt{1 - \alpha_1\alpha_2\alpha_3}\bar{\varepsilon}$, without appealing to induction.
- With $\alpha_s = 0.99$ for every s , find the smallest t with $\bar{\alpha}_t < 0.01$. What does x_t look like at that point?
- Derive $\mathbb{E}[x_0 \mid x_\sigma]$ for a **three**-point dataset $x_0 \in \{-1, 0, +1\}$ with equal priors. Is it still a $\tanh$?

- (f) For the two-point denoiser, compute $\text{Var}(x_0 \mid x_\sigma)$ and sketch it against x_σ . Where is the denoiser least certain, and does that match intuition?
- (g) Verify Tweedie's formula for the two-point mixture: show $x + \sigma^2 \nabla_x \log p_\sigma(x) = \tanh(x/\sigma^2)$.
- (h) Derive the Gaussian KL formula in d dimensions for $\Sigma_1 = \Sigma_2 = \sigma^2 I$ and confirm it is $\|\mu_1 - \mu_2\|^2 / 2\sigma^2$.
- (i) Compute $\text{KL}(\mathcal{N}(0, 1) \parallel \mathcal{N}(\mu, 1))$ and $\text{KL}(\mathcal{N}(\mu, 1) \parallel \mathcal{N}(0, 1))$. Why are these two equal when the variance case was not?
- (j) A network is trained with squared error to predict ε from x_t . Using §7.1, state exactly what function it converges to, and then use $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon$ to rewrite that function in terms of $\mathbb{E}_{x_0|x_t}[x_0]$. Say explicitly which distribution each expectation in your answer is taken over.